

ToolCallLimitMiddleware

This control the agent by limiting the number of tool calls, either globally or specific tools. Useful for:

- Preventing excessive calls to expensive external APIs
- Limiting web searches or database queries
- Enforcing rate limits on specific tool usage
- Protecting against runaway agent loops

```
from langchain.agents import create_agent
from langchain.agents.middleware import ToolCallLimitMiddleware

agent = create_agent(
    model="gpt-5.4",
    tools=[search_tool, database_tool],
    middleware=[
        # Global limit
        ToolCallLimitMiddleware(thread_limit=20, run_limit=10),
        # Tool-specific limit
        ToolCallLimitMiddleware(
            tool_name="search",
            thread_limit=5,
            run_limit=3,
        ),
    ],
)
```

- `thread_limit` = maximum tool calls across all runs in a thread (conversation) = the tool can be used at most 20 times total across the entire conversation, no matter how many messages you send
- `run_limit` = maximum tool calls per single invocation (one user message -> response cycle). reset with each new user message. =if `run_limit` =10 so if you ask one question, it can call tools up to 10 times to response. New message = back to 10 again
 - `None` = no limit